Congenital Myopathy With Reduced Type 2 Muscle Fibers (CMYO14 / MYL1-related congenital myopathy)

MONDO:0034109 · OMIM:618414 · Causal gene: *MYL1* (HGNC:7582)

Summary (Answer to the Research Question)

Congenital myopathy with reduced type 2 muscle fibers — formally **congenital myopathy-14 (CMYO14)** — is an **ultra-rare, autosomal-recessive, severe neonatal-onset skeletal muscle disease caused by biallelic loss-of-function variants in *MYL1***, the gene encoding the **fast-twitch (type II) skeletal-muscle myosin essential/alkali light chain (ELC; isoforms MLC1F/MLC3F)**. Loss of functional fast ELC destabilizes the myosin motor specifically in type II fibers, producing arrested fast-fiber development/maturation with marked **type II fiber hypotrophy or absence** on muscle biopsy, while type I (slow) fibers — which use a different ELC, MYL3 — are relatively spared. Clinically this manifests as profound **congenital hypotonia, generalized weakness, respiratory insufficiency, and feeding difficulty**, and the disorder has a severe prognosis. Fewer than ~6 patients have been reported worldwide (first described by Ravenscroft et al. 2018,).

P 30215711

This report was assembled from **disease-level aggregated resources (OMIM, gnomAD, NCBI Gene/HomoloGene, HGNC/UniProt/Ensembl)** and from the **small set of primary case reports** that constitute the entire human literature; it is not derived from EHR cohorts (no registry exists for such a rare disease).

1. Disease Information

- **What it is:** A congenital myopathy defined molecularly by MYL1 deficiency and histologically by selective smallness/atrophy or absence of fast-twitch (type II) myofibres. "Congenital myopathies are typically characterised by early onset hypotonia, weakness and hallmark features on biopsy" (Ravenscroft 2018, 30215711).
- **Key identifiers:**
- **MONDO:** MONDO:0034109
- **OMIM phenotype:** 618414 (Myopathy, congenital, 14; CMYO14)
- **OMIM gene:** *MYL1* 160780
- **Mondo synonyms / OMIM title:** "Myopathy, congenital, with fast-twitch (type II) fiber atrophy" / "…fibre atrophy"
- **Orphanet:** Listed under congenital myopathies; no separate large epidemiology entry (ultra-rare).
- **ICD-11:** 8C70 (Primary disorders of muscles / congenital myopathy) — approximate mapping; **ICD-10:** G71.2 (congenital myopathies).
- **MeSH:** "Myopathies, Structural, Congenital" (D020914); gene MeSH "Myosin Light Chains."
- **Gene identifiers:** HGNC:7582; NCBI Gene 4632; Ensembl ENSG00000168530; UniProt **P05976**; locus **2q34**.
- **Synonyms/aliases:** CMYO14, CMYP14, MYL1-related congenital myopathy, myopathy with fast-twitch type II fiber atrophy; gene aliases MLC1F, MLC3F, MLC1/3, MYOFTA.
- **Source type:** Disease-level/aggregated + individual case reports (not EHR).

Ontology: MONDO:0034109.

2. Etiology

- **Primary cause — genetic:** Biallelic (homozygous or compound-heterozygous) **loss-of-function variants in *MYL1***. "We identified variants in the gene (MYL1) encoding the skeletal muscle fast-twitch specific myosin essential light chain (ELC) in both probands" (Ravenscroft 2018,

).

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It is a monogenic Mendelian disorder; no environmental or infectious cause.

- **Genetic risk factors:** The only risk factor is inheriting two pathogenic *MYL1* alleles. **Consanguinity** is a major contributor (multiple reported families are consanguineous; homozygous variants). No susceptibility loci or GWAS associations (too rare).
- **Environmental risk factors:** None known. Not toxin/lifestyle/occupation-related.
- **Protective factors:** A **single functional *MYL1* allele is fully protective** — carriers are asymptomatic (see gnomAD constraint, §9). No dietary/lifestyle protective factors are relevant.
- **Gene–environment interactions:** None documented; disease is fully genetically determined.

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3. Phenotypes

All features have **congenital/neonatal onset** and are **severe**; course is early and often life-threatening. Frequencies are qualitative given ≤6 patients.

Phenotype	Type	HPO term	Onset / severity / frequency
Severe generalized hypotonia ("floppy infant")	clinical sign	HP:0001290 (neonatal hypotonia HP:0001319)	Congenital; severe; ~all patients
Generalized muscle weakness	clinical sign	HP:0003324	Congenital; severe; ~all
Respiratory insufficiency / ventilator dependence	clinical sign	HP:0002093 / HP:0004887	Neonatal; severe; most patients
Feeding difficulties / need for nutritional support	symptom	HP:0011968	Neonatal; severe; most
Facial weakness / myopathic facies	clinical sign	HP:0010628 / HP:0000338	Congenital; variable
Skeletal anomalies / contractures (incl. arthrogryposis-like)	physical	HP:0002804 / HP:0001371	Congenital; variable
Craniofacial/dysmorphic anomalies	physical	HP:0001999	Reported in newest case (Barington 2026, P 41568526)
Reduced fetal movement / antenatal onset	symptom	HP:0001558	Prenatal in some
Type II (fast) fiber hypotrophy/atrophy/absence on biopsy	laboratory/ pathology	HP:0003554 (type II muscle fiber atrophy)	Universal defining feature

Key quotes: "Both individuals showed a very severe congenital myopathy, characterised by congenital hypotonia and weakness, requiring ventilatory and nutritional assistance" (Madrigal/Natera-de Benito 2025, P 40488356); "a neonate with severe hypotonia, respiratory insufficiency, and skeletal anomalies... the patient presented with craniofacial anomalies not previously described" (Barington 2026, P 41568526).

Quality of life: Profound — affected neonates typically require respiratory support (invasive/non-invasive ventilation) and tube feeding, with major impairment of all gross motor function. Formal QoL instruments (EQ-5D/PROMIS) have not been applied given rarity and age.

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4. Genetic / Molecular Information

- **Causal gene:** *MYL1* (myosin light chain 1), OMIM 160780, HGNC:7582, locus **2q34**, Ensembl ENSG00000168530, UniProt **P05976**. Encodes the **non-phosphorylatable alkali/essential light chain of fast skeletal muscle myosin**; "This gene encodes a myosin alkali light chain expressed in fast skeletal muscle. Two transcript variants have been identified" (NCBI RefSeq). The two isoforms **MLC1F and MLC3F** arise by alternative promoter/splicing and are **type II–fiber specific**.
- **Pathogenic variants reported (all germline, biallelic):**
- **c.479-2A>G** — essential splice-acceptor, predicted exon-5 skipping (homozygous, Proband 1; Ravenscroft 2018,

  P 30215711

) — **Pathogenic**.
- **c.488T>G, p.(Met163Arg)** — missense, predicted to impair binding to the IQ domain of myosin heavy chain and to destabilize the protein (homozygous, Proband 2;

  P 30215711

) — **Pathogenic/Likely pathogenic**.
- **Three further novel variants** across two individuals (Madrigal/Natera-de Benito 2025,

  P 40488356

).
- **c.479-25T>C** — deep-intronic splice variant; RNA from patient muscle showed inclusion of 19 bp of intron 4 → frameshift/premature stop in most transcripts, plus in-frame exon 4–5 skipping in a minority (homozygous; Barington 2026,

  P 41568526

).
- **Variant classes:** splice-site, missense, and deep-intronic splice-altering. No recurrent/founder allele; all private.

- **Allele frequency:** All pathogenic alleles are absent/ultra-rare in gnomAD. General MYL1 LoF is **tolerated** in population (obs 16 vs exp 21.8 LoF; see §9).
- **Functional consequence: Loss of function** (null/unstable protein) — "MYL1 was markedly reduced in skeletal muscle from both probands, suggesting that the missense substitution likely results in an unstable protein"
  (P 30215711).
  Not gain-of-function or dominant-negative.
- **Modifier genes:** None identified. (Hypothetically, expression of other fast ELC/regulatory light chains could modulate severity, but this is unproven.)
- **Epigenetics:** No disease-specific DNA-methylation/histone findings reported.
- **Chromosomal abnormalities:** None — point/splice variants only; no CNV/structural etiology reported.

Ontology/annotation: HGNC:7582; UniProt P05976; GO molecular function "structural constituent of muscle" (GO:0008307), "myosin light chain binding."

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5. Environmental Information

- **Environmental factors:** None. Purely genetic Mendelian disorder.
- **Lifestyle factors:** Not applicable.
- **Infectious agents:** None (not infectious or immune-mediated).

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6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic *MYL1* loss-of-function variant** (splice/missense/deep-intronic) **results in** aberrant/absent *MYL1* mRNA (e.g., exon skipping, frameshift + premature stop) — *demonstrated by patient RNA studies*
   (P 41568526).

2. This **leads to** absent or unstable **fast-skeletal essential/alkali light chain (MLC1F/MLC3F)** protein — *demonstrated by markedly reduced MYL1 on Western blot of patient muscle* (PMID 30215711).
3. Loss of ELC **results in** a **destabilized myosin lever arm/neck** and impaired heavy-chain–light-chain coupling in **type II (fast) fibers** — *inferred from in-vitro motility: removing myosin light chains reduces actin sliding velocity ~11-fold (8.8→0.8 μm/s) without reducing ATPase* (Lowey 1993, PMID 8413589).
4. This **leads to impaired force transduction and defective fast-twitch myofibre development/maturation** (type II fibers remain small and retain **developmental myosin**) — *demonstrated histologically* (PMID 40488356).
5. The branch point: **type I (slow) fibers are relatively spared** because they use a **different ELC gene (MYL3)**; they remain normal-to-hypertrophic, producing the **"floret" pattern** (small type II fibers clustered around larger type I fibers) (PMID 40488356).
6. Fiber-level dysfunction **results in generalized weakness and hypotonia**, affecting respiratory and bulbar musculature → **respiratory insufficiency and feeding difficulty** — *the clinical endpoint* (PMID 30215711 PMID 40488356 PMID 41568526).
7. In vivo validation: **zebrafish myl1 knockdown results in** disrupted muscle structure and impaired touch-evoked (fast-twitch–dependent) escape responses, confirming the chain (PMID 30215711).

Mechanistic detail by category

- **Molecular pathways / biochemical process:** Actin–myosin cross-bridge cycling and the sarcomeric contractile apparatus (Reactome "Striated Muscle Contraction," R-HSA-390522).

The defect is structural-contractile, **not** a classic signaling cascade (no Wnt/MAPK/mTOR involvement implicated).

- **Protein dysfunction:** Loss of function via protein instability/absence; the missense p.Met163Arg is predicted to "impede intermolecular interactions that facilitate binding to the IQ domain of myosin heavy chain, thus likely impacting on the structure and functioning of the myosin motor"
(30215711).
- **Cellular processes:** Arrested fast-fiber differentiation/maturation (persistent developmental/neonatal myosin); dystrophic-like/myopathic changes on biopsy, but **not** a primary apoptotic/inflammatory/autophagic disease.
- **Metabolic changes:** None primary (not a metabolic myopathy).
- **Immune involvement:** None.
- **Tissue damage:** Secondary dystrophic-like change without primary necrosis/fibrosis as the driver.
- **Molecular profiling:** Patient-muscle **RNA analysis** (splicing), **Western blot** (protein loss), **immunohistochemistry/immunofluorescence/EM** (fiber typing, developmental myosin, ultrastructure)
(40488356 , 41568526).
No large transcriptomic/proteomic/metabolomic datasets exist.

Ontology: GO biological process — skeletal muscle contraction (GO:0003009), myofibril assembly (GO:0030239), muscle filament sliding (GO:0030049), skeletal muscle fiber development (GO:0048741); GO cellular component — myosin complex (GO:0016459), myofibril (GO:0030016), sarcomere/A band. CL — fast muscle cell (CL:0000190), skeletal muscle fiber (CL:0008002).

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7. Anatomical Structures Affected

- **Organ level: Skeletal muscle** is the primary affected organ (UBERON:0001134). Secondary/system involvement: **respiratory system** (ventilatory failure from weak respiratory/diaphragmatic muscles, UBERON:0001004), **digestive/feeding** (bulbar weakness →

dysphagia), **skeletal system** (contractures, craniofacial/skeletal anomalies). **No primary cardiac involvement** is characteristic (cardiac myosin uses MYL3/MYL4, not MYL1), and brain/CNS structure is intrinsically normal (weakness is myopathic, not neurogenic).

- **Tissue/cell level: Striated (skeletal) muscle tissue**; the targeted population is the **fast-twitch type II skeletal muscle fiber** (CL:0000190 fast muscle cell). Type I/slow fibers (CL:0000189) are relatively spared.
- **Subcellular level: Sarcomere/myofibril** — specifically the **myosin thick filament / myosin complex** (GO:0016459) at the head–neck (lever-arm) region; **A-band**. No primary mitochondrial, nuclear, ER, or lysosomal pathology.
- **Localization / lateralization: Generalized and bilateral/symmetric**, affecting axial, limb, respiratory, facial and bulbar muscles.

Ontology: UBERON:0001134 (skeletal muscle tissue), UBERON:0001103 (diaphragm), UBERON:0001004 (respiratory system).

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8. Temporal Development

- **Onset: Congenital** — antenatal (reduced fetal movements in some) to **neonatal**; onset pattern is **present-at-birth/early**, not acute-episodic.
- **Progression:** The myopathy itself is best characterized as **early-severe and largely static-to-slowly-evolving** (a congenital myopathy rather than a relentlessly degenerative dystrophy), but the **clinical course is dominated by severe early respiratory/feeding compromise**; too few long-surviving patients exist to define a late natural history. Duration: **chronic/lifelong**.
- **Patterns:** No remission. **Critical period** = **neonatal/infantile**, where respiratory and nutritional support is decisive for survival.

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9. Inheritance and Population

- **Inheritance: Autosomal recessive.** "CMYO14… an ultrarare autosomal recessive disorder caused by biallelic variants in MYL1" (Barington 2026,

P 41568526

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- **Penetrance:** Complete in biallelic carriers (all reported biallelic individuals affected); **heterozygotes unaffected**.
- **Expressivity:** Uniformly severe to date, with some phenotypic expansion (craniofacial anomalies).
- **Anticipation:** Not applicable (not a repeat-expansion disorder).
- **Germline mosaicism / founder effect:** None reported; all variants private; several **consanguineous** pedigrees → homozygosity.
- **gnomAD constraint (computational):** pLI = 7.2×10^{-5}; **oe_lof = 0.735 (90% CI 0.498–1.117)**; obs_lof = 16 vs exp 21.8 → **MYL1 is NOT loss-of-function-intolerant / not haploinsufficient**, consistent with recessive inheritance and asymptomatic carriers.
- **Carrier frequency:** Not formally established; expected low given private alleles (higher within consanguineous communities).
- **Epidemiology: Ultra-rare — ≤6 patients reported worldwide (2018–2026).** True prevalence/incidence are unknown (likely <1 per million). No sex predilection; reported in consanguineous families of varied ancestry.

Ontology/inheritance: HP:0000007 (autosomal recessive inheritance).

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10. Diagnostics

- **Laboratory:** Serum **creatine kinase** may be normal or mildly elevated (as in most congenital myopathies); no specific blood biomarker. No specific metabolite/enzyme assay.
- **Electrophysiology:** EMG typically shows a **myopathic pattern**; nerve conduction normal (distinguishes from neurogenic floppy infant, e.g., SMA).
- **Imaging:** Muscle MRI/ultrasound can show myopathic change (nonspecific); no pathognomonic pattern reported for this ultra-rare entity.
- **Muscle biopsy (key):** Dystrophic-like/myopathic changes with **hypotrophy/atrophy or absence of type II (fast) fibers**; small fibers arranged around larger type I fibers in a **"floret" pattern**; **persistent developmental myosin** expression; EM shows aberrant myofibrillar features. ATPase/myosin-isoform immunohistochemistry demonstrates the fiber-type disproportion; **Western blot shows markedly reduced MYL1 protein**

(P 30215711). P 40488356

- **Genetic testing (confirmatory): WES or WGS**, or a **congenital myopathy next-generation-sequencing gene panel** that includes *MYL1*, is the diagnostic approach of choice — "massive parallel sequencing" underlies modern diagnosis (P 40488356). **WGS and RNA/functional studies are specifically valuable for deep-intronic/splice variants** that WES or panels may miss — "underscores the diagnostic importance of intron variants, highlighting the value of combining in silico splice prediction with functional RNA analyses" (P 41568526). Single-gene *MYL1* testing is appropriate once biopsy points to type II fiber deficiency. CMA/karyotype/FISH/mtDNA/repeat testing are **not** indicated.
- **Clinical criteria / differential diagnosis:** No formal diagnostic criteria; diagnosis = clinical congenital myopathy + type II fiber deficiency on biopsy + biallelic *MYL1* variants. **Differential diagnosis:** other causes of the floppy infant — **SMA (SMN1)**; other congenital myopathies (nemaline/*ACTA1*, core/*RYR1*, centronuclear/*MTM1,DNM2,BIN1*); **congenital fiber-type disproportion** (classically **type I** fiber smallness — the mirror image of MYL1's type II smallness; P 26526626); congenital muscular dystrophies; metabolic/mitochondrial myopathies; congenital myasthenic syndromes. Biopsy fiber-typing plus genetics distinguish these (floppy-infant work-up: P 23743156).
- **Screening:** Not on newborn-screening panels; **cascade carrier testing** of relatives and **carrier screening** in consanguineous families once a familial variant is known.

Ontology/LOINC: CK (LOINC 2157-6); genetic testing NCIT.

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11. Outcome / Prognosis

- **Survival/mortality: Severe prognosis.** Reported patients have profound neonatal weakness with **ventilator and nutritional dependence**; given the very small, recent cohort, formal survival statistics are unavailable, but the phenotype is life-threatening in infancy. No 5-/10-year figures exist.
- **Morbidity/function:** Severe motor disability; dependence on respiratory and feeding support; major impact on all developmental milestones.
- **Complications: Respiratory failure and respiratory infections**, aspiration, failure to thrive, orthopedic complications (contractures, scoliosis) — typical of severe congenital myopathy.
- **Recovery potential:** No cure; supportive care can stabilize but not reverse. Congenital myopathies in general are static-to-slowly-progressive, so some patients who survive the neonatal period may achieve relative stability, but evidence is limited for MYL1.
- **Prognostic factors:** Degree of respiratory involvement and severity of the null genotype are the main determinants; no validated prognostic biomarker.

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12. Treatment

No disease-specific or curative therapy exists. Management is **supportive/multidisciplinary**, as for severe congenital myopathies:

- **Pharmacotherapy:** None disease-specific. No approved drug; no pharmacogenomic guidance. (General congenital-myopathy care avoids depolarizing agents where malignant-hyperthermia risk is a concern, but MH is a *RYR1* issue, not established for MYL1.)
- **Respiratory support (cornerstone):** Non-invasive or invasive **mechanical ventilation**, airway clearance (NCIT: Mechanical Ventilation C70909).
- **Nutritional support: Nasogastric/gastrostomy feeding** for bulbar weakness/failure to thrive (NCIT: Enteral Nutrition/Gastrostomy).
- **Rehabilitation: Physical therapy, occupational therapy, speech therapy**; contracture prevention (stretching/orthoses) (NCIT: Physical Therapy C15327, Occupational Therapy, Speech Therapy).
- **Surgical/orthopedic:** Management of contractures and **scoliosis** as needed.

- **Advanced therapeutics (experimental/none approved):** No gene therapy, ASO, cell therapy, or targeted therapy exists or is in trial for MYL1 specifically. Conceptually, because the mechanism is **recessive loss-of-function**, **gene replacement/augmentation** (e.g., AAV-delivered MYL1) or **splice-modulating ASOs** (for the intronic splice alleles) are rational future directions, but remain hypothetical. The zebrafish model provides a platform for preclinical testing (P 30215711).
- **Treatment outcomes:** Supportive care improves survival/comfort but does not alter the underlying deficit; no response-rate data.
- **Strategy:** Genotype-guided counseling; multidisciplinary neuromuscular care pathway.

13. Prevention

- **Primary prevention:** Not applicable to disease occurrence beyond reproductive genetics. **Genetic counseling** for carrier couples (25% recurrence risk per pregnancy).
- **Secondary prevention:** Early recognition of the floppy infant and prompt respiratory/nutritional support; no population screening.
- **Tertiary prevention:** Prevent complications — respiratory surveillance/immunizations against respiratory pathogens (e.g., influenza, RSV prophylaxis, pneumococcus) in the medically fragile infant; aspiration precautions; contracture/scoliosis prevention.
- **Genetic/reproductive options: Carrier (cascade) testing**, **prenatal diagnosis**, and **preimplantation genetic testing (PGT-M)** are available for families with a known biallelic genotype. Especially relevant in **consanguineous** families.
- **Immunization/public-health/environmental:** No disease-specific vaccine or environmental intervention (not infectious/environmental).

14. Other Species / Natural Disease

- **Taxonomy / orthologs (HomoloGene group 23194):** *MYL1* is highly conserved — **mouse *Myl1*** (NCBI Gene 17901; NCBITaxon:10090), **rat *Myl1*** (56781), **zebrafish *myl1*** (58143; NCBITaxon:7955), **chicken** (396470), **cattle** (317657), **dog** (478896), **frog** (394551). In mouse,

Myl1 is a canonical **fast-fiber marker** (expressed alongside Myh3/Myh8/Mylpf; Amann 2014,

25172430

).

- **Natural disease in animals: None documented** — no entry in OMIA for naturally occurring MYL1 myopathy; no companion-animal/wildlife counterpart or specific affected breed (no VBO breed). **No zoonotic potential / cross-species transmission** (genetic disorder).
- **Comparative biology:** Strong evolutionary conservation of fast-fiber ELC function supports translational validity of models; fiber-type specificity (fast ELC = MYL1; slow/cardiac ELC = MYL3) is conserved across vertebrates.

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15. Model Organisms

- **Validated model — zebrafish (*Danio rerio*, NCBITaxon:7955):** Morpholino ***myl1* knockdown** produced "abnormal morphology, disrupted muscle structure and impaired touch-evoked escape responses, thus confirming that skeletal muscle fast-twitch specific myosin ELC is critical for myofibre development and function" (Ravenscroft 2018,

30215711

). The touch-evoked escape readout is **fast-twitch dependent**, directly mirroring the human type II deficit — **good phenotype recapitulation** of the muscle-structural and fast-fiber-function phenotype. (Resource: ZFIN.)
- **Mammalian models: No dedicated *Myl1*-knockout mouse myopathy model** has been reported for this disease (gap); mouse *Myl1* is used mainly as a fast-fiber expression marker. IMPC/MGI (Gene MGI for *Myl1*) could host future alleles. **Limitation:** absence of a mammalian model limits preclinical therapeutic testing; morpholino knockdown has known transient/off-target caveats.
- **In vitro / cellular:** No established patient iPSC-myotube or immortalized line is reported specifically, though patient muscle (histology, Western blot, EM, RNA) has been studied directly
(

40488356

).

41568526

- **Applications:** Models are suited to studying fast-fiber myofibrillogenesis, ELC–myosin lever-arm mechanics, and (prospectively) gene-replacement or splice-correction therapy.

Supported vs Refuted Hypotheses

Supported: - Biallelic *MYL1* LoF causes CMYO14 (human genetics + zebrafish; P 30215711).
- The disease selectively affects **type II/fast fibers** because MYL1 is the fast-fiber-specific ELC, while type I fibers use MYL3 (histology + gene biology; P 40488356).
- Mechanism is **loss of function** (unstable/absent protein), with ELC required for normal myosin shortening velocity (P 8413589; P 30215711).
- Strictly **autosomal recessive** with asymptomatic carriers (gnomAD constraint; P 41568526).
- Deep-intronic splice variants expand the mutational spectrum and require RNA/WGS diagnostics (P 41568526).

Refuted / not supported: - Not haploinsufficient/dominant (gnomAD pLI≈0; carriers unaffected). - Not a metabolic, inflammatory, mitochondrial, or primary cardiac disease. - No environmental/infectious etiology; no founder effect established.

Limitations and Future Directions

- **Tiny evidence base:** ≤6 patients; frequencies, natural history, and survival are estimates, not statistics.

- **No mammalian genetic model** and **no clinical trials/therapies** — gene-replacement and splice-modulating ASO strategies are rational but untested.
- **Unknowns:** long-term prognosis in survivors; existence of milder/later-onset phenotypes; modifier genes; precise fraction of cardiac/respiratory muscle MYL1 dependence.
- **Priorities:** generate a conditional/null mouse or stable zebrafish mutant; patient iPSC-myotube models; registry/natural-history collection; evaluate AAV-MYL1 and ASO splice correction.

Key references

- Ravenscroft G, et al. *Bi-allelic mutations in MYL1 cause a severe congenital myopathy.* Hum Mol Genet / J Med Genet 2018.

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- Madrigal/Natera-de Benito D, et al. *MYL1-Related Congenital Myopathy: Clinical, Genetic and Pathological Insights.* 2025.

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- Barington M, et al. *Novel MYL1 Intron Variant With Expanded Phenotype.* 2026.

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- Sitbon YH, et al. *Insights into myosin regulatory and essential light chains…* 2020 (review).

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- Amann R, et al. (fast-fiber marker *Myl1* in mouse) 2014.

  P 25172430

- Kissiedu J, Prayson RA. *Congenital fiber type disproportion.* 2016 (differential).

  P 26526626

- Gene/constraint resources: NCBI Gene 4632; gnomAD (MYL1 constraint); HGNC:7582; UniProt P05976.
